

Temporal confidence aggregation. In the TAA task, frames closer to the end of a clip carry stronger causal evidence about an impending event, since the accident—if any—occurs shortly after the clip ends. Motivated by this *temporal saliency prior*, we aggregate the per-frame predictions into a single clip-level confidence anchor p_{final} by retaining the last-frame prediction, which empirically exhibits the highest signal-to-noise ratio among all frame positions.

Temporal risk prior reconstruction. Accident risk possesses a well-known *monotonicity prior*: the probability of an imminent collision increases monotonically as the temporal distance to the event decreases. We exploit this inductive bias to reconstruct a dense, physically plausible frame-level risk curve from the clip-level anchor. Specifically, we parameterise the reconstruction as

$$p_t = p_{\text{start}} + (\beta p_{\text{final}} - p_{\text{start}}) \cdot h(t/T; k, \mu), \quad (2)$$

where h is a monotonically increasing temporal basis function that blends exponential and logarithmic schedules, k and μ are shape parameters estimated from the second-order statistics of the raw curve, $\beta < 1$ is a head-room factor ensuring strict monotonicity below the end anchor, and p_{start} is the initial risk level. This step effectively *propagates* the model’s high-confidence late-segment signal backward along the temporal axis under a physically motivated constraint.

Prediction distribution alignment. Even after temporal reconstruction, the predicted confidence distribution on the target domain remains systematically shifted relative to the source domain’s decision boundary. Since no target-domain labels are available, we perform *unsupervised recalibration* using only the moments of the model’s own prediction distribution on the target set. Concretely, we compute the empirical median m of $\{p_{\text{final}}\}$ across all target clips as a domain-specific statistic, and apply an order-preserving mapping that aligns the predicted distribution to the expected decision boundary. This procedure is analogous to test-time batch normalisation in unsupervised domain adaptation: it shifts the activation statistics to match the target distribution without retraining. Crucially, because the only target-set quantity used is a moment of the model’s own output distribution, the zero-shot constraint is fully respected.

3. Experiments

3.1. Implementation Details

The full model (87.1 M parameters) is fine-tuned with AdamW [7], learning rate 1×10^{-5} for the backbone and 1×10^{-4} for the head, weight decay 5×10^{-4} , batch size 3 with 8-step gradient accumulation (effective batch 24), 20 epochs, StepLR ($\gamma = 0.1$ every 5 epochs), gradient clipping with max norm 5.0, and automatic mixed precision (AMP). Training takes approximately 17 hours on a single NVIDIA

V100 GPU. We select the submitted checkpoint based on the validation AP, AUC, and mTTA.

Model scale. The VideoMAE-v2 Base backbone has 86.2 M parameters; the per-frame head adds 0.9 M, totalling 87.1 M trainable parameters.

Test-time adaptation hyper-parameters. On the source validation set, the raw model produces median $p_{\text{final}} \approx 0.34$ and clip-mean ≈ 0.05 . The temporal risk prior uses $k \in [7, 15]$, $\mu \in [0.2, 0.8]$, and head-room factor $\beta = 0.998$. The distribution alignment step maps clips in $[m, 0.5)$ order-preservingly into $[0.505, 0.7]$.

3.2. Evaluation Protocol

We follow the official evaluation protocol of the CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1]. The evaluation measures detection quality via frame-level Average Precision (AP) and Area Under the ROC Curve (AUC), while early and stable risk awareness is assessed via Time-To-Accident (TTA) and Stable TTA (STTA). Higher AP and AUC indicate better detection capability, whereas larger TTA metrics reflect earlier anticipation. On our held-out Nexar [9] validation split, we report the AP and AUC metrics. Additionally, we present the competition’s official weighted composite score evaluated on the unseen test set, denoted as “Private LB” in our results.

3.3. Results

Beyond the final pipeline of Sec. 2, we ran two experiments to test design choices that, *a priori*, looked promising but turned out to hurt the score. Both are reported here because their negative results directly motivate decisions in our final method. In both, the *only* change from the final pipeline is the one explicitly stated; everything else is held constant.

Late-Window Classification. Instead of feeding the model a 16-frame window over the full 5 s clip, we extract only the *last 2 s* of every clip, uniformly sub-sample 16 frames from this trailing window, and produce a single clip-level binary decision (rather than a per-frame risk sequence). The clip-level probability is then replicated across all 150 output positions. Trained with the same backbone and class-balanced split as the proposed model, this variant reaches a validation AP = 0.66 and AUC = 0.69 (Tab. 1, row B), clearly below the proposed per-frame model. On the private leaderboard the gap widens to -0.027 with respect to our final submission. The underlying reason is that, by replicating a single scalar across all 150 positions, the late-window output has no temporal resolution by construction, so the post-processor of Sec. 2.4—which is order-preserving—cannot rebuild a rising risk curve from a flat one. Empirically, this confirms that *discarding the first 3 s of context costs more than the simpler late-time supervision saves*; we therefore